12TH CHEMISTRY · CHAPTER 23 · SOLVED EXERCISE

Carbonyl Compounds and Carboxylic Acids

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS:

I. Ketones are prepared by the oxidation of alcohols:

(a) Primary (b) Secondary
(c) Tertiary (d) Quaternary

Answer: (b) Secondary — Secondary alcohols are oxidised to ketones; primary alcohols give aldehydes.

II. Which of the following will not give the iodoform test?

(a) Acetaldehyde (b) 2-pentanone
(c) 3-Pentanoacid (d) Acetone

Answer: (c) 3-Pentanoacid — It has no $CH_3CO–$ group; the other three contain it.

III. When a carboxyl acid reacts with an alchol, the product is an:

(a) Ethers (b) Ester
(c) Anhydride (d) Amide

Answer: (b) Ester — This is esterification.

IV. Carboxylic acids react with carbonates to produce gas.

(a) H_2 (b) CO
(c) CO_2 (d) CH_4

Answer: (c) CO_2 — A salt, water and carbon dioxide are formed.

V. Which type of alcohols produces aldehydes?

(a) Primary (b) Secondary
(c) Tertiary (d) both a and b

Answer: (a) Primary — Primary alcohols are oxidised to aldehydes.

VI. The hybridization of the oxygen of carbonyl group is:

(a) sp (b) sp^2
(c) sp^3 (d) dsp^2

Answer: (b) sp^2 — Both carbon and oxygen of the carbonyl group are sp^2 hybridized.

VII. Which is the main product of the reaction between acetic acid and PCl_5.

(a) $CH_3–Cl$ (b) CH_3COCl
(c) HCl (d) $POCl_3$

Answer: (b) CH_3COCl — The –OH group is replaced by Cl to give ethanoyl chloride.

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VIII. Acidic character of chlorine substituted carboxylic acid is higher because chlorine is:

(a) Electropositive (b) Electron with drawing due to –I
(c) Electron donating due to +I (d) having three loan pairs

Answer: (b) Electron with drawing due to –I — It disperses the negative charge of the carboxylate ion and stabilises it.

IX. When formic acid is oxidized, which of the following is formed?

(a) water (b) carbon dioxide
(c) carbon dioxide and water (d) carbon monoxide

Answer: (c) carbon dioxide and water — $HCOOH + [O] \rightarrow CO_2 + H_2O$.

X. The mechanism by which acyl chlorides react with alcohols is.

(a) Addition (b) Elimination
(c) Addition-elimination (d) Substitution

Answer: (c) Addition-elimination — The alcohol adds to the carbonyl carbon and then HCl is eliminated.

MCQ II option (c) is printed as "3-Pentanoacid" and MCQ III as "carboxyl acid ... alchol"; they are reproduced as printed.

Q2. Attempt the following short answer questions:

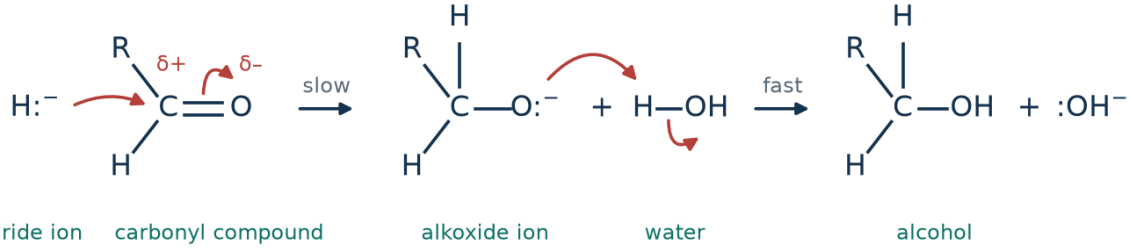
a) How is ethanol converted to acetic acid?

Ethanol is refluxed with excess acidified $K_2Cr_2O_7$ (or acidified $KMnO_4$). It is oxidised first to ethanal and then to ethanoic (acetic) acid.

$$CH_3\text{–}CH_2\text{–}OH + 2[O] \rightarrow CH_3\text{–}COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{ heat})$$

b) Give the mechanism for the reductions of ethanal with $NaBH_4$.

The reducing agent supplies the hydride ion, H^-, which is the nucleophile. It attacks the δ+ carbonyl carbon and the π electrons shift to oxygen, forming an **alkoxide ion**. The alkoxide ion then takes a proton from water to give the **alcohol** and a hydroxide ion.



For ethanal, R is the methyl group and the alcohol formed is ethanol:

$$CH_3\text{–}CHO + 2[H] \rightarrow CH_3\text{–}CH_2\text{–}OH \ (NaBH_4)$$

c) What is iodoform reaction?

It is the reaction of a compound containing the $CH_3CO–$ group (ethanal or a methyl ketone) with iodine in the presence of sodium hydroxide to form a yellow crystalline precipitate of triiodomethane (iodoform), CHI_3. It is used as a test for the $CH_3CO–$ group.

$$CH_3–CHO + 3I_2 + 4NaOH \rightarrow CHI_3\downarrow + HCOONa + 3NaI + 3H_2O$$

d) Methanoic acid can be oxidized but other carboxylic acids cannot. Why?

The molecule of methanoic acid, H–CO–OH, contains an aldehydic portion (–CHO) as well as the carboxylic portion. Because of this aldehydic portion it is oxidised by weak as well as strong oxidising agents (Tollens' reagent, Fehling's solution) to carbon dioxide and water. Other carboxylic acids have an alkyl group in place of this hydrogen, so they have no aldehydic portion.

e) Explain the influence of chlorine substitution on the acidity of carboxylic acids.

Chlorine is an electron-withdrawing atom (–I effect). It pulls electron density away from the carboxylate group, disperses the negative charge and stabilises the ion, so the acid ionises more. The more chlorine atoms present, the stronger the acid. Trichloroacetic acid is comparable in strength to some mineral acids.

Acid	Formula	pK_a
Acetic (ethanoic) acid	$CH_3–COOH$	4.74
Chloroacetic acid	$ClCH_2–COOH$	2.86
Dichloroethanoic acid	$Cl_2CH–COOH$	1.26
Trichloroacetic acid	$Cl_3C–COOH$	0.64

Acid strength: $Cl_3C–COOH > Cl_2CH–COOH > ClCH_2–COOH > CH_3–COOH$

f) Give the reaction of acyl chloride with phenol.

When an acyl chloride is warmed with phenol, a phenyl ester and hydrogen chloride are formed.

$$CH_3–COCl + C_6H_5–OH \rightarrow CH_3–COO–C_6H_5 \text{ (phenyl ethanoate)} + HCl$$

g) Explain why carboxylic acid is more acidic than ethanol.

In the ethanoate ion the negative charge is delocalised over the two oxygen atoms of the carboxylate group, so the ion is stabilised and is less likely to accept a proton again. In the ethoxide ion the charge stays on one oxygen atom and is intensified by the electron-donating ethyl group, so it is unstable. Hence ethanoic acid ionises much more than ethanol.

K_a: $RCOOH = 10^{-5}$ $R–OH = 10^{-16}$

h) Give reaction of methanal with the Tollen's reagent.

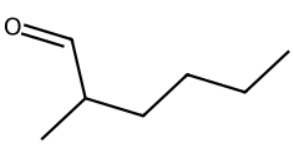
Methanal reduces the silver ions of Tollens' reagent to metallic silver, which forms a silver mirror on the wall of the test tube; methanal itself is oxidised to the methanoate ion.

$$HCHO + 2[Ag(NH_3)_2]OH \rightarrow HCOONH_4 + 2Ag\downarrow + 3NH_3 + H_2O$$

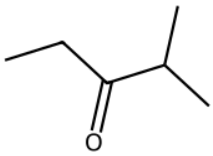
The textbook describes this test in words and gives an equation only for methanoic acid; the equation above is the standard one for an aldehyde.

i) Write down structural formulae of the following compounds. i) 2-methylhexanal ii) 2-methyl-3-pentanone iii) 3-hydroxypentanl iv) 4-nitro-2-pentanone v) 2-chlorobutanol

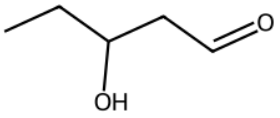
	Name	Structural formula
i)	2-methylhexanal	$CH_3–CH_2–CH_2–CH_2–CH(CH_3)–CHO$
ii)	2-methyl-3-pentanone	$(CH_3)_2CH–CO–CH_2–CH_3$
iii)	3-hydroxypentanal	$CH_3–CH_2–CH(OH)–CH_2–CHO$
iv)	4-nitro-2-pentanone	$CH_3–CO–CH_2–CH(NO_2)–CH_3$
v)	2-chlorobutanol	$CH_3–CH_2–CHCl–CH_2–OH$



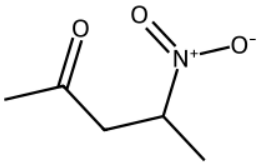
i) 2-methylhexanal



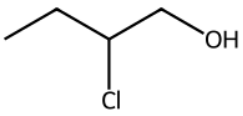
ii) 2-methyl-3-pentanone



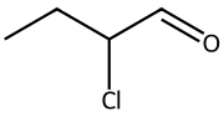
iii) 3-hydroxypentanal



iv) 4-nitro-2-pentanone



v) 2-chlorobutanol



(2-chlorobutanal)

Part iii) is printed as "3-hydroxypentanl" (3-hydroxypentanal). Part v) is solved as printed (2-chlorobutan-1-ol); if 2-chlorobutanal was intended, its formula is $CH_3–CH_2–CHCl–CHO$ (last structure).

Q3. CONSTRUCTED RESPONSE QUESTIONS:

a) Write down the equation for the reaction of CH_3CH_2-CHO with $LiAlH_4$ (dry ether).

$$CH_3–CH_2–CHO + 2[H] \rightarrow CH_3–CH_2–CH_2–OH\ (LiAlH_4, \text{dry ether})$$

Propanal is reduced to the primary alcohol propan-1-ol.

b) An alcohol, $C_4H_{10}O$ undergoes oxidation to form a carboxylic acid, $C_4H_8O_2$. Give a chemical equation along with conditions.

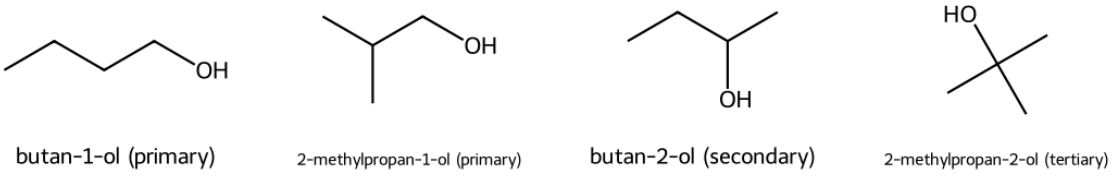
The alcohol must be a primary alcohol, e.g. butan-1-ol. It is refluxed with excess acidified $K_2Cr_2O_7$ (or acidified $KMnO_4$) and is oxidised to butanoic acid.

$$CH_3–CH_2–CH_2–CH_2–OH + 2[O] \rightarrow CH_3–CH_2–CH_2–COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{ reflux})$$

c) Write the structures of all the isomeric alcohols with formula $C_4H_{10}O$. Give their systematic names. Which one of them forms the above carboxylic acid? Explain why the others do not.

There are four isomeric alcohols:

Systematic name	Structure	Type	On oxidation (reflux)
butan-1-ol	$CH_3–CH_2–CH_2–CH_2–OH$	Primary	butanoic acid
2-methylpropan-1-ol	$(CH_3)_2CH–CH_2–OH$	Primary	2-methylpropanoic acid
butan-2-ol	$CH_3–CH_2–CH(OH)–CH_3$	Secondary	butanone (a ketone)
2-methylpropan-2-ol	$(CH_3)_3C–OH$	Tertiary	not oxidised



butan-1-ol (primary) 2-methylpropan-1-ol (primary) butan-2-ol (secondary) 2-methylpropan-2-ol (tertiary)

Butan-1-ol forms the acid of part (b), butanoic acid. Butan-2-ol is a secondary alcohol and gives only a ketone; 2-methylpropan-2-ol is a tertiary alcohol and is not oxidised. 2-Methylpropan-1-ol is also a primary alcohol; it gives 2-methylpropanoic acid, which is a different acid with the same formula $C_4H_8O_2$.

d) Propanoic acid can be reduced to propan-1-ol by reduction with $LiAlH_4$. Give a balanced chemical equation. Can we use $NaBH_4$ for this reaction? If not, give reason.

$$CH_3–CH_2–COOH + 4[H] \rightarrow CH_3–CH_2–CH_2–OH + H_2O \ (LiAlH_4)$$

No. $NaBH_4$ (sodium tetrahydridoborate(III)) is not a powerful enough reducing agent; it will not reduce carboxylic acids. Only the powerful reducing agent $LiAlH_4$ can do so.

DESCRIPTIVE QUESTIONS:

Q4. What are aldehydes and ketones? Give the preparation of each.

Aldehydes and ketones are the classes of compounds that contain the carbonyl group, >C=O. In aldehydes the carbonyl carbon is attached to at least one hydrogen atom (R–CHO); in ketones it is attached to two alkyl groups (R–CO–R).

Preparation of aldehydes:

Aldehydes are prepared by the oxidation of primary alcohols with acidified $K_2Cr_2O_7$ or acidified $KMnO_4$.

$$CH_3–CH_2–OH + [O] \rightarrow CH_3–CHO + H_2O \ (K_2Cr_2O_7 + H_2SO_{4(aq)}, \text{heat})$$

In the preparation of ethanal, the acidified $K_2Cr_2O_7$ is added dropwise to warm ethanol. The aldehyde is **distilled off as soon as it is formed**; this prevents its further oxidation to ethanoic acid. The temperature is kept above the boiling point of ethanal but below that of ethanol, since ethanol has the higher boiling point. Pure ethanal is obtained by redistillation.

Preparation of ketones:

Ketones are prepared by the oxidation of secondary alcohols, by the same procedure. Since ketones are not converted into carboxylic acids under these conditions, the ketone is not distilled off immediately.

$$CH_3–CH(OH)–CH_3 + [O] \rightarrow CH_3–CO–CH_3 + H_2O \ (K_2Cr_2O_7 + H_2SO_4, \text{heat})$$

Q5. Give four methods for the preparation of carboxylic acids.

a) Oxidation of primary alcohols and aldehydes: the starting material is refluxed with excess of either acidified $KMnO_4$ or acidified $K_2Cr_2O_7$.

$$CH_3–CH_2–OH + 2[O] \rightarrow CH_3–COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{heat})$$
$$CH_3–CHO + [O] \rightarrow CH_3–COOH \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{heat})$$

b) Hydrolysis of alkane nitriles: nitriles are directly hydrolysed to carboxylic acids in acidic medium. In alkaline hydrolysis they first give salts of carboxylic acids, which yield the acids on acidification.

$$CH_3–CN + 2H_2O + HCl \rightarrow CH_3–COOH + NH_4Cl$$
$$CH_3–CN + H_2O + NaOH \rightarrow CH_3–COONa + NH_3\uparrow$$
$$CH_3–COONa + HCl \rightarrow CH_3–COOH + NaCl$$

c) Hydrolysis of esters: when an ester is refluxed with dilute acid or dilute alkali, it is hydrolysed to the carboxylic acid or the carboxylate ion.

$$CH_3–COO–CH_2–CH_3 + H_2O_{(l)} \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH \ (H_3O^+)$$
$$CH_3–COO–CH_2–CH_3 + NaOH_{(aq)} \rightarrow CH_3–COONa + CH_3–CH_2–OH$$

$$CH_3–COONa + HCl \rightarrow CH_3–COOH + NaCl$$

d) Preparation of benzenecarboxylic acid (benzoic acid): methylbenzene is readily oxidised by acidified $KMnO_4$ or $K_2Cr_2O_7$ to benzoic acid.

$$C_6H_5–CH_3 + 3[O] \rightarrow C_6H_5–COOH + H_2O \ (KMnO_4, \text{dil. } H_2SO_4, \text{heat})$$

Q6. How does ethanoic acid reacts with: I) NaOH ii) ethanol iii) SOCl iv) PCl_5

I) NaOH: neutralisation takes place; sodium ethanoate and water are formed.

$$CH_3–COOH + NaOH \rightarrow CH_3–COONa + H_2O$$

ii) Ethanol: on heating with ethanol in the presence of concentrated sulphuric acid, the ester ethyl ethanoate is formed (esterification).

$$CH_3–COOH + C_2H_5OH \rightleftharpoons CH_3–COOC_2H_5 + H_2O \ (\text{conc. } H_2SO_4)$$

iii) $SOCl_2$: the –OH group is replaced by chlorine, giving ethanoyl chloride.

$$CH_3–COOH + SOCl_2 \rightarrow CH_3–COCl + HCl + SO_2$$

iv) PCl_5: ethanoyl chloride is formed.

$$CH_3–COOH + PCl_5 \rightarrow CH_3–COCl + HCl + POCl_3$$

Part iii) is printed as "SOCl"; the reagent is thionyl chloride, $SOCl_2$.

Q7. Give the mechanisms of the reactions of carbonyl compounds with: i) HCN b) $NaBH_4$

Both are nucleophilic addition reactions.

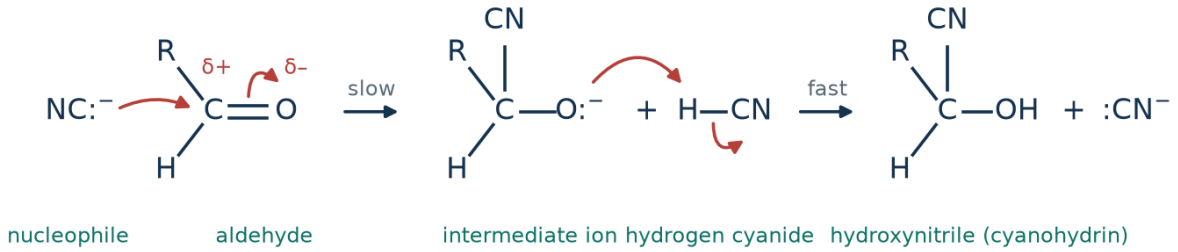
i) HCN:

HCN is neither a nucleophile (there is no lone pair on its carbon atom) nor a strong acid that ionises to a reasonable extent, so it does not produce enough nucleophile (CN^-) to start the reaction. Therefore a slight excess of KCN is used, so that some free CN^- ions are present. These ions start the reaction and catalyse it.

$$\text{i) } KCN \text{ (excess)} + HCl \text{ (limiting)} \rightarrow HCN + KCl$$

ii) The nucleophile CN^- attacks the δ+ carbonyl carbon; the π electrons move to the oxygen, giving an intermediate ion with a negative charge on oxygen.

iii) This ion takes a proton from HCN to form the hydroxyalkanenitrile, and a CN^- ion is regenerated.

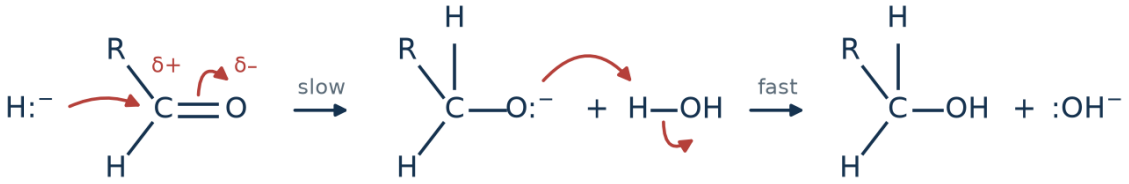


nucleophile aldehyde intermediate ion hydrogen cyanide hydroxynitrile (cyanohydrin)

Although both steps are reversible, the equilibria lie far to the right, so most carbonyl compounds give high yields of cyanohydrins.

b) $NaBH_4$:

The reducing agent supplies the hydride ion, H^-, which is the nucleophile. It attacks the δ+ carbonyl carbon and the π electrons shift to oxygen, forming an **alkoxide ion**. The alkoxide ion then takes a proton from water to give the **alcohol** and a hydroxide ion.



hydride ion carbonyl compound alkoxide ion water alcohol

Q8. Explain the relative acidities of chlorine substituted carboxylic acids.

The strength of a carboxylic acid depends on the stability of its carboxylate ion. An electron-withdrawing group (–I effect) stabilises the ion by dispersing its negative charge and increases the acidity.

Chlorine is an electron-withdrawing atom (–I effect). It pulls electron density away from the carboxylate group, disperses the negative charge and stabilises the ion, so the acid ionises more. The more chlorine atoms present, the stronger the acid. Trichloroacetic acid is comparable in strength to some mineral acids.

Acid	Formula	pK_a
Acetic (ethanoic) acid	CH_3–COOH	4.74
Chloroacetic acid	$ClCH_2$–COOH	2.86
Dichloroethanoic acid	Cl_2CH–COOH	1.26
Trichloroacetic acid	Cl_3C–COOH	0.64

Acid strength: Cl_3C–COOH > Cl_2CH–COOH > $ClCH_2$–COOH > CH_3–COOH

Q9. Explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes.

The reactivity of acyl chlorides, alkyl chlorides and aryl chlorides can be compared by the ease of their hydrolysis.

$CH_3–COCl + H_2O \rightarrow CH_3–COOH + HCl$ (room temperature, rapid)

$CH_3–CH_2–Cl + H_2O \rightarrow CH_3–CH_2–OH + HCl$ (heat under reflux)

$C_6H_5–Cl + H_2O \rightarrow$ no reaction

Compound	Reason	Ease of hydrolysis
Ethanoyl chloride (acyl chloride)	The carbonyl carbon has a large partial positive charge due to the strong electron-withdrawing effect of two highly electronegative atoms, oxygen and chlorine, attached directly to it. Nucleophilic attack of H_2O is very fast.	Complete within a few minutes at room temperature
Chloroethane (alkyl chloride)	The carbon gets a comparatively smaller partial positive charge because it is attached to only one chlorine atom. Nucleophilic attack of water is slow.	Takes 14 days even at 100 °C
Chlorobenzene (aryl chloride)	The lone pair on chlorine is delocalised with the π electrons of the benzene ring. This gives double-bond character to the C–Cl bond, making it so strong that it does not break.	No reaction

Ease of hydrolysis: acyl chloride > alkyl chloride > aryl chloride

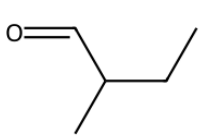
QUICK CHECK QUESTIONS (IN-CHAPTER)

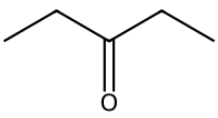
These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

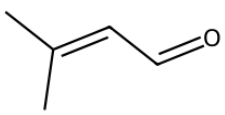
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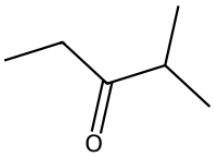
a) Draw the structures for the following according to IUPAC system. i) 2-,methylbutanal ii) pentan-3-one iii) 3-methylbutenal d) 4-methylpentan-3-one

	Name	Condensed structure
i)	2-methylbutanal	$CH_3–CH_2–CH(CH_3)–CHO$
ii)	pentan-3-one	$CH_3–CH_2–CO–CH_2–CH_3$
iii)	3-methylbutenal (3-methylbut-2-enal)	$(CH_3)_2C=CH–CHO$
d)	4-methylpentan-3-one	$CH_3–CH_2–CO–CH(CH_3)_2$


i) 2-methylbutanal


ii) pentan-3-one

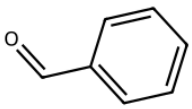

iii) 3-methylbut-2-enal

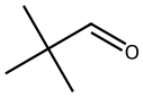

d) 2-methylpentan-3-one

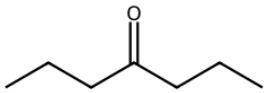
The fourth part is lettered d) in the textbook. The name "4-methylpentan-3-one" does not use the lowest numbers; the structure it describes is correctly named 2-methylpentan-3-one.

b) Name the following compounds (IUPAC) system). i) C_6H_5–CHO ii) $(CH_3)_3C$–CHO iii) $(CH_3CH_2CH_2)_2C=O$

i) benzaldehyde (benzenecarbaldehyde) **ii)** 2,2-dimethylpropanal **iii)** heptan-4-one


i) benzaldehyde


ii) 2,2-dimethylpropanal


iii) heptan-4-one

c) Rank the following compounds in order of increasing reactivity towards nucleophilic attack. i) $(CH_3)_3CCOCH_3$ ii) $(CH_3)_3CCOC(CH_3)_3$ iii) $(CH_3)_2CO$

$(CH_3)_3CCOC(CH_3)_3$ (ii) < $(CH_3)_3CCOCH_3$ (i) < $(CH_3)_2CO$ (iii)

The bulkier and the more electron-releasing the alkyl groups on the carbonyl carbon, the greater the steric hindrance and the smaller its positive charge, so the lower the reactivity.

Quick Check 23.2

a) If compound A has the molecular formula C_3H_6O, it reacts with $NaBH_4$ to give compound B, C_3H_8O. Both A and B give a yellow precipitate with I_2/NaOH. Identify both compounds (A & B).

A gives the iodoform test, so it contains the CH_3CO– group: **A is propanone, CH_3–CO–CH_3**. Its reduction gives a secondary alcohol with the CH_3–CH(OH)– group, which also gives the test: **B is propan-2-ol, CH_3–CH(OH)–CH_3**.

b) Why methanol does not give iodoform test while ethanol does?

Alcohols having the CH_3–CH(OH)– group also give the iodoform test. Ethanol gives the test, while methanol does not: ethanol has the CH_3–CH_2–O structure, which acts like a methyl ketone structure (it is oxidised to CH_3–C=O), so it can form iodoform. Methanol has no such group.

c) Describe a test (a different one in each case) that you could use to distinguish between the following pairs of compounds. i) CH_3CH_2CHO and CH_3COCH_3 ii) $CH_3CH_2COCH_2CH_3$ and $CH_3CH_2CH_2COCH_3$ iii) CH_2=CH–CH_2–OH and CH_3CH_2CHO

i) Tollens' test: warm each with ammoniacal silver nitrate. Propanal (an aldehyde) gives a silver mirror; propanone does not.

ii) Iodoform test: warm each with I_2 and NaOH. Pentan-2-one (a methyl ketone) gives a yellow precipitate of CHI_3; pentan-3-one does not.

iii) 2,4-DNPH test: propanal (a carbonyl compound) gives an orange crystalline solid; the alcohol prop-2-en-1-ol does not.

An extra "CH_3CH_2CHO" is printed after the word "compounds." in the stem of this question.

Quick Check 23.3

a) Write the equation for the oxidation of propanol with acidified $K_2Cr_2O_7$?

$$CH_3–CH_2–CH_2–OH + 2[O] \rightarrow CH_3–CH_2–COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{ reflux})$$

Propan-1-ol is oxidised through propanal to propanoic acid.

b) What products you expect from following? Give the chemical equations. I) Hydrolysis of CH_3CH_2CN ii) Oxidation of ethylbenzene iii) Hydrolysis of ethyl ethanoate

I) Propanoic acid (and ammonium chloride in acidic medium):

$$CH_3–CH_2–CN + 2H_2O + HCl \rightarrow CH_3–CH_2–COOH + NH_4Cl$$

ii) Benzoic acid; the whole side chain is oxidised to –COOH:

$$C_6H_5–CH_2–CH_3 + 6[O] \rightarrow C_6H_5–COOH + CO_2 + 2H_2O \ (KMnO_4, \text{ dil. } H_2SO_4, \text{ heat})$$

iii) Ethanoic acid and ethanol:

$$CH_3–COO–CH_2–CH_3 + H_2O \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH \ (H_3O^+)$$

The textbook gives the oxidation of methylbenzene only; the ethylbenzene equation is written on the same pattern.

Quick Check 23.4

a) Methanoic acid is stronger than ethanoic acid. Give reason.

In ethanoic acid the methyl group is electron-donating (+I effect). It intensifies the negative charge on the ethanoate ion and destabilises it, which decreases the acidity. Methanoic acid has only a hydrogen atom in place of the methyl group, so its ion is more stable and it is about 10 times stronger.

b) Place the following acids in order of strength, starting with the strongest acid first. i) CH_3CH_2COOH ii) CH_3CCl_2COOH iii) $CH_3CHClCOOH$

$$CH_3CCl_2COOH \text{ (ii)} > CH_3CHClCOOH \text{ (iii)} > CH_3CH_2COOH \text{ (i)}$$

Chlorine is electron-withdrawing (–I effect) and stabilises the carboxylate ion; two chlorine atoms have a greater effect than one.

c) Explain why ethanoic acid is a stronger acid than ethanol.

In the ethanoate ion the negative charge is delocalised over the two oxygen atoms of the carboxylate group, so the ion is stabilised and is less likely to accept a proton again. In the ethoxide ion the charge stays on one oxygen atom and is intensified by the electron-donating ethyl group, so it is unstable. Hence ethanoic acid ionises much more than ethanol.

Quick Check 23.5

a) Why do phenols not react directly with carboxylic acids to form phenyl esters?

In phenol a lone pair of the oxygen atom is delocalised into the benzene ring, so phenol is a poor nucleophile and cannot attack the less reactive carbonyl carbon of a carboxylic acid. Phenyl esters are therefore made from the much more reactive acyl chlorides.

This reason is not stated directly in the textbook; it follows from the delocalisation of the oxygen lone pair in phenol (Chapter 22) and the high reactivity of acyl chlorides.

b) Phenyl ethanoate is the ester responsible for the fragrance of peach and strawberry. It can be prepared from the ethanoyl chloride and phenol. Write down a chemical reaction for the preparation of this ester.

$$CH_3\text{–}COCl + C_6H_5\text{–}OH \rightarrow CH_3\text{–}COO\text{–}C_6H_5 + HCl$$

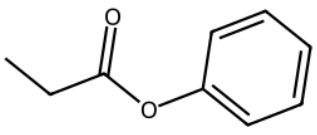
c) Predict the products of the following reactions, write the chemical equations. I) Chlorobenzene and ethanol ii) Propanoyl chloride and phenol

I) No reaction. The C–Cl bond of chlorobenzene has double-bond character and is too strong to be broken by a nucleophile.

$$C_6H_5\text{–}Cl + C_2H_5\text{–}OH \rightarrow \text{no reaction}$$

ii) Phenyl propanoate and hydrogen chloride:

$$CH_3\text{–}CH_2\text{–}COCl + C_6H_5\text{–}OH \rightarrow CH_3\text{–}CH_2\text{–}COO\text{–}C_6H_5 + HCl$$



phenyl propanoate